Growing Hypergraphs with Homophily

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There are many extant models of hypergraphs with interactions governed by some form of homophily or assortativity, but almost all of these assume independence between edges conditional on node parameters. Relaxing this strong assumption, we study a mechanistic model of growing hypergraphs in which edge formation is influenced by the realization of previous edges and is tuned by homophily between binary node attributes. Edges form in this model as noisy copies of previous edges, where the transmission of nodes from one edge to the next depends on the labels. We derive a power law for the degree distribution in this model and describe the long-term dynamics of the joint distribution of labels contained in edges. This model defines a likelihood over a labeled hypergraph, allowing us to use standard maximum-likelihood techniques to structure algorithms. We estimate the model parameters on synthetic and real data via (stochastic) expectation maximization, allowing us to statistically quantify the signature of homophily in empirical polyadic systems. We also demonstrate an approach to community detection via simulated annealing which, though computationally expensive, achieves competitive results on certain data sets known to be challenging to standard community detection techniques derived from models that assume edge independence. Our findings highlight the benefits of moving beyond edge-independence assumptions in modeling and data analysis, and point to several directions of future work.

TODO LIST

PC: Must make parameter names consistent between text and figures.

PC: Standardize on "external" vs. "extant".

I. INTRODUCTION

Hypergraphs provide a natural, flexible data representation for complex systems with polyadic (multi-way) interactions, such as social interactions (both in-person [2] and online [3]), academic co-authorship [4], and ecological systems [5]. Much recent work has emphasized the importance of faithfully representing polyadic structure rather than reducing it to dyadic interactions [6, 7], though recent work has shown that some purportedly higher-order dynamics can be faithfully represented on dyadic networks under appropriate models [8].

Much attention in the theory of (hyper)graph mining has focused on developing novel techniques for community detection (also often called *clustering* or *partitioning*). Many but not all of these methods can be viewed as exact or approximate inference algorithms for stochastic generative models. A stochastic generative model is a parametrized probability distribution defined over hypergraphs. In one standard framework, a set $\mathcal{V}$ of n nodes is initialized along with a vector $\mathbf{z} \in [k]^n$ that assigns to each node one of k community labels. One then forms the edge set $\mathcal{E}$ iteratively by visiting each subset $R \subseteq \mathcal{V}$ and independently assigning a_R (possibly repeated) edges to R according to a distribution $p(a_R|\mathbf{z}_R, \theta)$, where $\mathbf{z}_R$ is the vector of labels of the nodes in R and θ is a set of model parameters. This results in a probability distribution over hypergraphs which can be factorized into a product of edge probabilities conditioned on the node labels and model parameters:

$$p(\mathcal{H}|\mathbf{z}, \theta) = \prod_{R \subseteq \mathcal{V}} p(a_R|\mathbf{z}_R, \theta)\ . \qquad (1)$$

Many generative models for community detection fall into this category, including hypergraph stochastic blockmodels [9–11] and some generalizations [12, 13]. The conditional independence property eq. (1) is extremely useful for computation, as factorizability of the likelihood enables many efficiencies in various inference algorithms, including approximate maximum-likelihood estimation via direct optimization [9] or expectation-maximization [10]; spectral methods [14–16]; and message passing [17]. Another recent optimization-based approach [18] is equivalent to a microcanonical version of the stochastic blockmodel, which satisfies an asymptotic form of edge-independence in the large sparse limit. Even a recent latent-space model [19] treats edges as independent samples, albeit from a very differently stochastic generating process.

Edge independence, however, is a strong and often unrealistic assumption for modeling hypergraph data. A parallel line of work has found that polyadic data sets often contain significant edge correlations [3, 20, 21], typically in the form of *similarity between interactions*. When an interaction occurs on a set $R \subseteq \mathcal{V}$, this increases the likelihood of observing a new interaction on a set $S \subseteq \mathcal{V}$ correlated with R in the sense that $R \cap S$ is larger than would be expected by chance. These overlaps can be shown to effect the kinds of dynamics which can unfold on the hypergraph [22, 23]. Several generative models for the growth of hypergraphs with edge-dependence have been proposed [24–28]. Several of these models express dependence through a copying or recombination mechanism, in which nodes collected from one or more existing hyperedges are gathered and noisily copied into a new edge as a block.

In this paper, we introduce a model of binary node-labeled hypergraphs in which edges form via a noisy copying interaction which depends the labels. We refer to this model as Copying Hyperedges Influenced by Latent Labels (CHILL). Our work is organized as follows. In Section II, we describe CHILL and derive limiting descriptions of the joint distribution of node labels within a given edge, as well as the degree distribution of the hypergraph. Evaluating the likelihood of our model requires marginalizing over a set of latent variables governing the noisy copy mechanism, which is computationally expensive. To estimate our model's parameters when node labels are known, we therefore describe a stochastic expectation-maximization algorithm in which each iteration has sublinear cost in the number of edges for sparse hypergraphs. We also describe a community detection algorithm for estimating node labels via simulated annealing on the CHILL likelihood. In Section IV we demonstrate our algorithms on synthetic and empirical data. We show that, on synthetic data sampled from CHILL, our stochastic expectation maximization algorithm is both consistent and scalable, and that our simulated annealing algorithm for community detection, though computationally expensive, outperforms other standard methods in certain parameter regimes. On empirical data, our parameter estimatates yield insight into the mechanisms governing edge formation Our community detection achieves competitive recovery of observed metadata labels on some data sets known to be challenging to other community detection methods. We conclude with some discussion of future directions for models of node-labeled hypergraphs with edge-dependence.

II. CHILL MODEL DESCRIPTION

The Copying Hyperedges Influenced by Latent Labels (CHILL) is a generative model for growing hypergraphs with binary node labels. CHILL is an extension of the Hyperedge Copy Model (HCM) of He et al. [26]. In their HCM, edges form through a combination of three distinct steps: a copy step, in which a previous edge is sampled and imperfectly copied; an extant node addition step, in which new nodes are added to the new edge from the existing node set; and a new node addition step, in which new nodes are added to the new edge from outside the existing node set. CHILL replicates these mechanisms but adds label-aware parameters to regulate each.

Formally, a CHILL state at time t is a hypergraph $\mathcal{H}^{(t)} = \left(\mathcal{V}^{(t)}, \mathcal{E}^{(t)}\right)$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of hyperedges, as well as a binary label vector $\mathbf{z} \in \{0,1\}^{|\mathcal{V}^{(t)}|}$ which assigns to each node a label of either 0 or 1. We let $\bar{z} = 1 - z$ denote the opposite label of z. At timestep t, we add edge e to $\mathcal{E}^{(t)}$ to obtain $\mathcal{E}^{(t+1)}$, where e is formed according to the following three-step process:

- **Edge-Copying**: An edge f is sampled uniformly at random from $\mathcal{E}^{(t-1)}$, and a node u is sampled uniformly at random from f. Then, each node v in f is included in e with probability ρ_+ if $z_v = z_u$ and with probability ρ_- if $z_v \neq z_u$.

- **Extant Node Addition**: We sample X_{z_u} nodes from $\mathcal{V} \setminus f$ with label z_u and add them to e, where X_z is a Poisson random variable with mean γ_+. We similarly sample $X_{\bar{z}_u}$ nodes from $\mathcal{V} \setminus f$ with label $\bar{z}_u$ and add them to e, where $X_{\bar{z}_u}$ is a Poisson random variable with mean γ_-.

- **Novel Node Addition**: We sample Y_{z_u} new nodes with label z_u and add them to e, where Y_z is a Poisson random variable with mean η_+. We similarly sample $Y_{\bar{z}_u}$ new nodes with label $\bar{z}_u$ and add them to e, where $Y_{\bar{z}_u}$ is a Poisson random variable with mean η_-.

Both our simulations and our statistical inference techniques (described below) were carried using the XGI package for polyadic data analysis in Python [29].

Figure 1 illustrates role of the model parameters in tuning the community structure of realized hypergraphs. We measure community structure by computing Newman-Girvan modularity [30] on the clique-projection of the hypergraph. For fixed γ_-, η_+, and η_-, increasing the difference $\rho_+ - \rho_-$ leads to higher modularity, as does increasing γ_+. Hypergraphs sampled from models with high values of ρ_+ and γ_+ tend to have label-homogeneous cores of densely-overlapping hyperedges, which are weakly connected to one another by edges with more heterogeneous label compositions. Hypergraphs with lower values of ρ_+ and γ_+ tend to have less-structured distributions of labels over hyperedges.

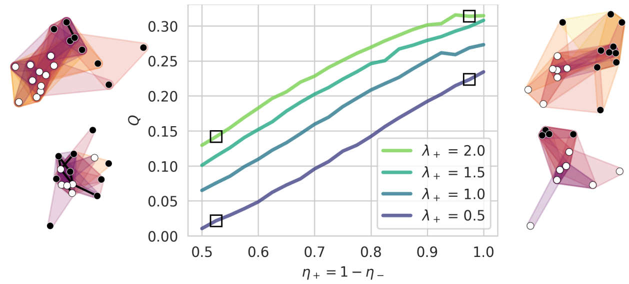

FIG. 1. Suite of model simulations illustrating the dependence of community structure, as measured by clique-projection modularity Q, on the CHILL parameters. The parameters $\gamma_- = 0.5$ $\eta_+ = 0.2$, and $\eta_- = 0.1$ are held fixed. Each line corresponds to a different value of γ_+ indicated by the legend. Along the horizontal axis, we vary ρ_+ and set $\rho_- = 1 - \rho_+$. Each parameter combination is simulated 10^2 times for 10^3 timesteps. On either side of the plot we show small hypergraphs sampled from parameter combinations indicated by the square markers.

A. Asymptotic Properties

1. *Joint Distribution of Labels in Edges*

Many asymptotic properties of our CHILL model can be described in terms of the joint distribution $p_{K_0,K_1}(k_0, k_1)$ of nodes of labels 0 and 1 in a uniformly random edge. This joint distribution can be obtained as the leading eigenvector of a linear map, as we now describe. Let $p^{(t)}_{K_0,K_1}(k_0, k_1)$ be the joint distribution of label counts in a uniformly random edge after t timesteps. Then, the probability $q(k'_0, k'_1)$ of realizing an edge with k'_0 nodes of label 0 and k'_1 nodes of label 1 at time $t+1$ is given by

$$q(k'_0, k'_1) = \sum_{k_0,k_1} T(k'_0, k'_1 | k_0, k_1) p^{(t)}_{K_0,K_1}(k_0, k_1) \ ,$$

where $T(k'_0, k'_1|k_0, k_1)$ gives the probability that the edge e realized in step $t+1$ has k'_0 nodes of label 0 and k'_1 nodes of label 1 given that the edge f sampled to form e has k_0 nodes of label 0 and k_1 nodes of label 1. At stationarity, we must have $q(k'_0, k'_1) = p^{(t)}_{K_0,K_1}(k'_0, k'_1) = p^{(t+1)}_{K_0,K_1}(k'_0, k'_1)$, so the stationarity condition can be obtained by solving the linear system

$$p^{(t+1)}_{K_0,K_1}(k'_0, k'_1) = \sum_{k_0,k_1} T(k'_0, k'_1 | k_0, k_1) p^{(t)}_{K_0,K_1}(k_0, k_1) \ .$$

The stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ is therefore given by the leading eigenvector of the doubly-indexed matrix T with entries $T(k'_0, k'_1|k_0, k_1)$. The entries of this matrix can be computed analytically. Conditional on choosing an edge f with k_0 nodes of label 0 and k_1 nodes of label 1, a node of label z is selected as the seed node u with probability $\frac{k_z}{k_0+k_1}$. Then, the number of nodes of labels z and $\bar{z}$ in e have distributions

$$\begin{aligned} k'_z &\sim 1 + \mathrm{Binom}\left(k_z - 1, \rho_+\right) + \mathrm{Poisson}(\gamma_+ + \eta_+) \ , \\ k'_{\bar{z}} &\sim \mathrm{Binom}\left(k_{\bar{z}}, \rho_-\right) + \mathrm{Poisson}(\gamma_- + \eta_-) \ . \end{aligned}$$

The probability of obtaining k'_0 nodes of label 0, conditioning on k_0 and k_1, is

$$\begin{aligned} t(k'_0|k_0, k_1) =& \frac{k_0}{k_0+k_1} \sum_{i=0}^{k'_0-1} \binom{k_0-1}{i} \rho_+^i (1-\rho_+)^{k_0-1-i} e^{-(\gamma_+ + \eta_+)} \frac{(\gamma_+ + \eta_+)^{k'_0-1-i}}{(k'_0-1-i)!} + \\ & \frac{k_1}{k_0+k_1} \sum_{i=0}^{k'_0} \binom{k_0}{i} \rho_-^i (1-\rho_-)^{k_0-i} e^{-(\gamma_- + \eta_-)} \frac{(\gamma_- + \eta_-)^{k'_0-i}}{(k'_0-i)!} \ . \end{aligned}$$

A parallel expression describes $t(k'_1|k_0, k_1)$, and the transition matrix entries are given by the product

$$T(k'_0, k'_1|k_0, k_1) = t(k'_0|k_0, k_1) t(k'_1|k_0, k_1) \ .$$

In Figure 2, we study the dynamics of our CHILL model as described by the matrix T. For fixed values of the parameters γ_+, γ_-, η_+ and η_-, the spectral gap $1 - \lambda_2$ of T is highest when ρ_- is large and ρ_+ is small. High values of the spectral gap correspond to fast convergence to the stationary distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ given by the Perron eigenvector $\mathbf{v}_1$ of T with eigenvalue 1 (panel (b)). We instead illustrate an alternative regime of relatively slow convergence in which ρ_+ is large and ρ_- is small. In this regime, the simulation may maintain transient dynamics in the direction of the second eigenvector $\mathbf{v}_2$ of T with eigenvalue $\lambda_2 < 1$ (panel (c)) for relatively long periods of time before converging to the stationary distribution. We illustrate this behavior in panel (d), where the system converges relatively slowly to the leading eigenvector $\mathbf{v}_1$ and away from the transient component $\mathbf{v}_2$. The resulting final simulation state matches the stationary distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ relatively closely (panel (e)).

Figure 2 shows that, when ρ_+ is sufficiently large, the stationary distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ is concentrated on edges in which typical edges have large majorities of nodes with the same label (panel b), thus displaying a

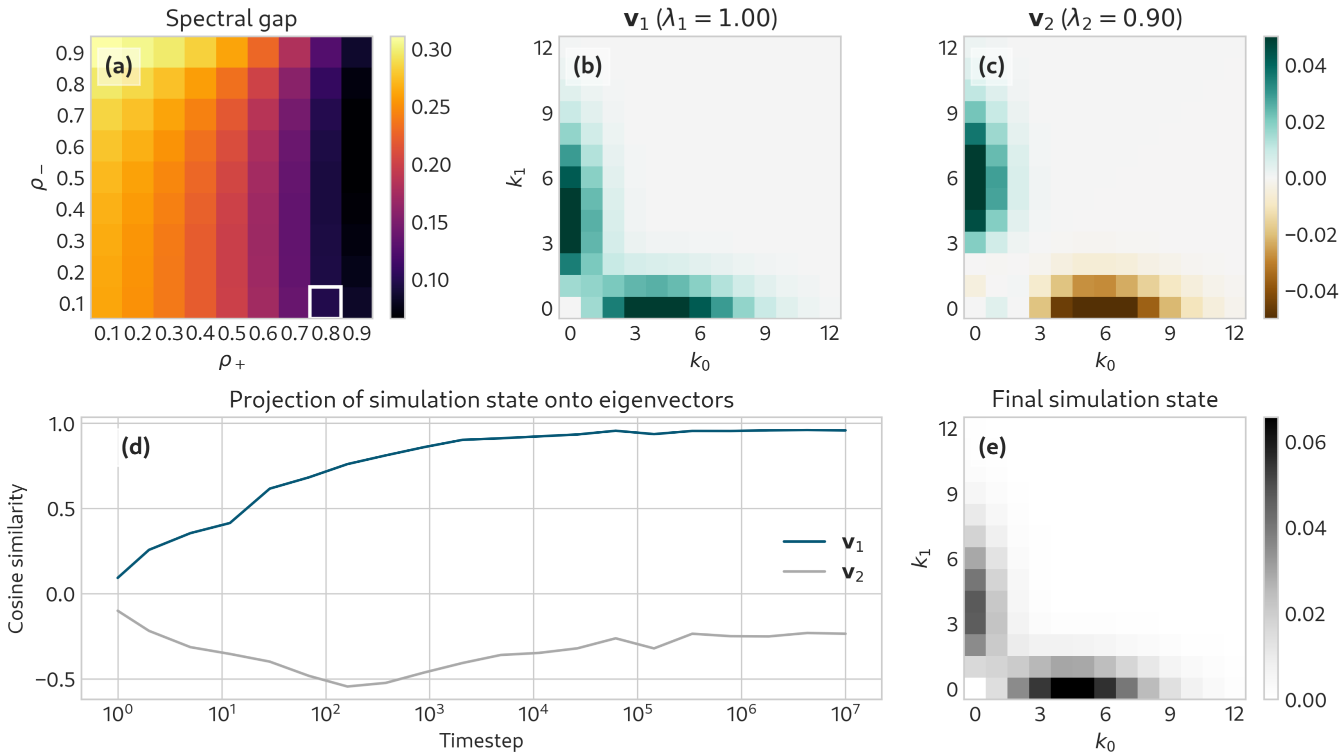

FIG. 2. Visualization of the long-run dynamics of our model, for parameters. (a): Spectral gap $1 - \lambda_2$ of the linear map T as a function of ρ_+ and ρ_- for fixed parameter values $\gamma_+ = 0.5$, $\gamma_- = \eta_-3$, $\eta_+ = 0.4$ and $\eta_- = 0.2$. The highlighted values $\rho_+ = 0.8$ and $\rho_- = 0.1$ are used in the remainder of the visualization. (b): Leading eigenvector $\mathbf{v}_1$ of T, describing the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ of label counts in edges. (c): Second eigenvector $\mathbf{v}_2$ of T. The transients of the model dynamics are dominated by the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$. (d): Euclidean projections of the simulated joint distribution $p^{(t)}_{K_0,K_1}(k_0, k_1)$ onto the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ during a simulation run with 10^7 timesteps. At each time point, we aggregate the most recent 1,000 edges to compute the joint distribution in order to approximate an "instantaneous" simulation state. (e): Empirical joint distribution $p^{(t)}_{K_0,K_1}(k_0, k_1)$ aggregated across 10^7 timesteps.

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form of higher-order homophily [31]. The second eigenvector v_2 (panel c), however, is antisymmetric, describing a transient in which one a single label group tends to dominate most edges. When ρ_+ is sufficiently large, the spectral gap $1 - \lambda_2$ is small (panel a), indicating that this transient decays slowly and that the system may be dominated by edges with a single majority label for relatively long periods of time before equilibrating to the leading eigenvector (panels d and e). This behavior suggests that, although the CHILL model ultimately converges to a stationary state with symmetry between the two node labels, it can also serve through its transients as a reasonable generative model of systems with label asymmetry.

2. *Degree Distribution*

The degree distribution of a hypergraph generated from the CHILL model follows an asymptotic power law with an exponent which can be computed in closed form from the model parameters and the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$. Our derivation is similar to that of He et al. [26], which in turn is based on a derivation by Mitzenmacher [32]. The argument requires a mean-field assumption: when a node u is selected from edge f in the edge-sampling step, the degree d of u is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e.

Under this assumption, it is possible to show that the degree distribution follows a power law with exponent

$$\zeta = 1 + \frac{\langle k \rangle}{1 + \rho_+(\rho_{00} + \rho_{11} - 1) + 2\rho_- \rho_{01}} \tag{2}$$
$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle .$$

In the expression for ρ_{ij} (and for $\langle k \rangle$) the expectation is taken with respect to the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$. Here, η can be interpreted as the expected number of novel nodes added per new edge, while σ can be interpreted as the expected number of nodes copied into a new edge from the sampled edge, normalized by the average degree of a node in the hypergraph. A derivation of this result and computational illustration are provided in Section A and Figure 7.

III. MODEL INFERENCE

In this section, we consider the problems of *parameter inference* (learning $\boldsymbol{\theta}$ given $\mathcal{H}$ and $\mathbf{z}$) and *community detection* (learning $\mathbf{z}$ given $\mathcal{H}$ and $\boldsymbol{\theta}$).

A. Parameter Estimation via Stochastic EM

We first consider the problem of learning the parameter vector $\boldsymbol{\theta} = (\rho_+, \rho_-, \gamma_+, \gamma_-, \eta_+, \eta_-)$ given an observed hypergraph $\mathcal{H}$ with sequential edge indices and known node labels $\mathbf{z}$. We aim to do this via maximum-likelihood estimation. Computing the complete data likelihood requires marginalizing over the model latent variables, which in this case are the edge f and node u selected at each timestep of the growth model to form the new edge e. For $e \in \mathcal{E}$, let t_e denote the time at which e was added to the hypergraph. We say that $f \prec e$ if $t_f < t_e$. Then, the complete data likelihood can be written

$$\mathbb{P}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta}) = \prod_{e\in\mathcal{E}} \sum_{f \prec e} \sum_{u \in f} p(e|f, u, \mathbf{z}, \boldsymbol{\theta}) p(f, u|\mathcal{E}_{t-1}, \mathbf{z}, \boldsymbol{\theta}) . \tag{3}$$

For notational simplicity, we have repressed the additional entries of $\mathbf{z}$ introduced by the novel node addition step. The requirement to sum over the latent variables at each timestep implies a computational cost to evaluate $\mathbb{P}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta})$ (or its derivatives) which is quadratic in the number of edges, making direct optimization of the likelihood expensive for data sets of even modest size. Batch expectation-maximization [33] is an alternative approach which maximizes the likelihood without explicitly computing eq. (3), but even this approach requires forming a belief distribution over the values of all latent variables at each timestep, which again incurs quadratic cost.

We therefore use a stochastic expectation-maximization (SEM) algorithm [34] to estimate the parameters $\boldsymbol{\theta}$. Each iteration of the SEM algorithm consists of two steps: the E-step, in which exponentially-weighted moving averages of the sufficient statistics are calculated using the current parameter estimates, and the M-step, in which the parameter estimates are updated using the expected sufficient statistics.

To initialize the SEM algorithm, we initialize a vector $\mathbf{s}^{(0)}$ of sufficient statistics to arbitrary values; in our model there is a sufficient statistic for each parameter. We then update a moving average of $\mathbf{s}$ as follows. In each algorithmic step ℓ, we pick an edge e uniformly at random from $\mathcal{H}$. Then, given current estimates $\hat{\boldsymbol{\theta}}^{(\ell)}$, we form a belief distribution over the latent identities of $f \in \mathcal{E}$ and $u \in f$ which could have been selected in the edge-copying step to form e. Given e, the probability that f and u were selected can be computed in closed form. Let $m^{(t-1)}$ denote the size of the edge set at time $t-1$.

We have

$$p(f, u|\mathcal{E}^{(t-1)}, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) = \frac{1}{|f|\, m^{(t-1)}} \triangleq r_{t-1}(f) ,$$

so Bayes' rule gives

$$p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)}) = \frac{p(e|f, u, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) r_{t-1}(f)}{\sum_{f' \prec e} \sum_{u' \in f'} p(e|f', u', \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) r_{t-1}(f')} . \tag{4}$$

A helpful note for computation is that $p(e|f, u, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) = 0$ if either $u \notin e \cap f$ or $f \not\prec e$. We then form the vector of expected sufficient statistics

$$s' = \sum_{f \prec e} \sum_{u \in f \cap e} p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)}) \sigma(e, f, u, \mathbf{z}) , \tag{5}$$

where $\sigma(e, f, u, \mathbf{z})$ is the vector of sufficient statistics for the parameters $\boldsymbol{\theta}$ given e, f, u, and $\mathbf{z}$. The entries of $\sigma(e, f, u, \mathbf{z})$ can be computed as functions of the node labels in e and f; we supply explicit formulae in Section B. We then perform the update

$$\mathbf{s}^{(\ell+1)} = (1 - \alpha^{(\ell)})\mathbf{s}^{(\ell)} + \alpha^{(\ell)}\mathbf{s}' ,$$

where $\alpha^{(\ell)}$ is the learning rate at step ℓ; we use an exponentially decaying learning rate in our experiments. Our estimate of the parameter vector $\boldsymbol{\theta}$ is then given by a function $\hat{\boldsymbol{\theta}}^{(\ell+1)} = g(\mathbf{s}^{(\ell+1)})$ of the sufficient statistics, which we also supply in Section B. Algorithm 1 summarizes the steps of this algorithm. Our version of the algorithm uses an online computation of $\mathbf{s}'$, which allows us to perform the double-sum appearing in eqs. (4) and (5) only once, and without explicitly storing the resulting belief distribution at any point.

Algorithm 1 CHILL SEM Update Step

Require: $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, $\mathbf{z}$, $\hat{\boldsymbol{\theta}}^{(\ell)}$, $\mathbf{s}^{(\ell)}$

```
e ← Uniform(ℰ)
a ← 0
b ← 0
for f ∈ ℰ do
    if f ≺ e then
        for u ∈ e ∩ f do
            a ← a + p(f, u|e, θ̂^(ℓ)) σ(e, f, u, z)
            b ← b + p(f, u|e, θ̂^(ℓ))
s' ← a/b
s^(ℓ+1) ← (1 − α^(ℓ)) s^(ℓ) + α^(ℓ) s'
θ̂^(ℓ+1) ← g(s^(ℓ+1))
return θ̂^(ℓ+1), s^(ℓ+1)
```

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PC: Made revisions. Happy with it but need a small comment to say why 800 (and why not 100, or 1,000, which would have been obvious default choices).

We perform SEM by repeating Algorithm 1 until a stopping criterion is met. We define the function

$$h(\hat{\boldsymbol{\theta}}^{(\ell)}, \hat{\boldsymbol{\theta}}^{(\ell')}) = \min_i \left\{ \left| \hat{\theta}_i^{(\ell)} - \hat{\theta}_i^{(\ell')} \right| - \epsilon_{\text{rel}} - \epsilon_{\text{abs}} \hat{\theta}_i^{(\ell')} \right\} .$$

The function h here can be viewed as combining absolute and relative error tolerances, enabling the algorithm to declare convergence even in cases when the parameter estimates are near zero. In our implementation, we declare convergence at timestep ℓ when $h(\hat{\boldsymbol{\theta}}^{(\ell)}, \hat{\boldsymbol{\theta}}^{(\ell')}) \geq 0$ for all ℓ' in the most recent 800 iterations of the algorithm. In our experiments we used $\epsilon_{\text{rel}} = 0.01$ and $\epsilon_{\text{abs}} = 0.015$.

B. Community Detection via Simulated Annealing

We also consider the community detection problem of learning the node labels $\mathbf{z}$ given an observed hypergraph $\mathcal{H}$ and a fixed parameter vector $\boldsymbol{\theta}$.

Equation (3) defines a likelihood function over both the parameter vector $\boldsymbol{\theta}$ and the node labels $\mathbf{z}$. This allows us to attempt to also learn $\mathbf{z}$ via maximum-likelihood estimation. Unlike in stochastic blockmodel variants in which there are several heuristics with favorable computational properties [9, 10, 13, 14, 16, 18, 35], the dependence structure encoded by CHILL makes it difficult to apply standard efficient heuristics; the development of such heuristics would be an important direction of future work. In this study, we therefore attempt to learn $\mathbf{z}$ via simulated annealing [36], which is a general-purpose optimization algorithm that can be used to approximately maximize the likelihood over $\mathbf{z}$. Our primary aim is to demonstrate that likelihood maximization in this model can detect label structure in at least some circumstances where more standard community detections may fail, thereby motivating the development of more efficient optimization heuristics in future work.

A direct evaluation of eq. (3) requires $O(m^2)$ terms, making it expensive to evaluate even for data sets of modest size. An important computational limitation of simulated annealing in this context is that the label z_i of node i appears in *all* terms of the likelihood eq. (3) corresponding to edges after which node i arrives. This includes edges in which node i does not participate, implying that we must perform $O(m^2)$ computations even to evaluate the change in likelihood under a proposed change to z_i.

To mitigate the computational burden, we use an intersection-based approximation to the likelihood of forming edge e in which we only sum over candidate copied edges f such that $f = \operatorname{argmax}_{f' \in \mathcal{E}} |f' \cap e|$. To justify this approximation, we start by observing that, if f is the edge sampled to form e, then the number of extant nodes required to form e is $x = |e| - |e \cap f| - |e \setminus \mathcal{V}|$. We'll argue that, when the number of nodes n is large, $p(e|f, u, \mathbf{z}, \boldsymbol{\theta}) \in O(n^{-x})$, which implies that edges f which do not maximize $|f \cap e|$ will have negligible contributions to the likelihood. The essence of the argument is that forming x extant nodes requires x independent samples from the sets $\mathcal{V}_0 \setminus f$ and $\mathcal{V}_1 \setminus f$, each of which have size scaling linearly with n. So, each independent sample has probability $O(n^{-1})$, and the probability of forming a given set of x extant nodes is $O(n^{-x})$. We detail this argument in Section C, obtaining an approximate objective function for simulated annealing

$$\tilde{\mathcal{L}}(\mathbf{z}) = \log \tilde{\mathbb{P}}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta}) ,$$

where $\tilde{\mathbb{P}}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta})$ is the likelihood of $\mathcal{H}$ and $\mathbf{z}$ under the intersection-based approximation. To run simulated annealing for a single epoch, we perform n individual update steps. In each step, we propose a change to the labels $\mathbf{z}$ by generating a node index j uniformly at random (without replacement) and evaluating the change in likelihood of "flipping" the binary node label ($z_j \leftarrow \bar{z}_j$).

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If the proposal increases $\tilde{\mathcal{L}}(\mathbf{z})$, it is accepted. If the proposal decreases $\tilde{\mathcal{L}}(\mathbf{z})$, however, we may also accept it with an acceptance probability $a(\ell)$ that decreases with the epoch ℓ. It is possible to store computational components required to evaluate $\tilde{\mathcal{L}}(\mathbf{z})$ in a data structure that allows us to compute the change in likelihood from the prior state without recomputing the entire likelihood. This results in a roughly 30-fold reduction in compute time on our empirical data sets at the expense of increasing memory requirements.

We use a custom acceptance schedule $a(\ell)$ for our simulated annealing implementation. Our custom approach is motivated by the observation that, in our application, the change in the objective $\tilde{\mathcal{L}}$ of a proposal can vary significantly depending on the size of the hypergraph under study. Thus, in our early experiments, traditional annealing schedules such as harmonic or exponential decay struggled to still occasionally accept proposals that decrease $\tilde{\mathcal{L}}$. Our custom implementation instead takes an adaptive approach based on the distribution of proposals previously observed. In the first epoch, we accept *all* proposals, while storing the changes the objective $\tilde{\mathcal{L}}$ produced by those proposals. At the end of the epoch, we compute the standard deviation $\sigma(\Delta\tilde{\mathcal{L}})$ of the changes in $\tilde{\mathcal{L}}$ observed during the epoch. For the remaining epochs, we accept proposals that decrease $\tilde{\mathcal{L}}$ by an amount $\Delta\tilde{\mathcal{L}}$ stochastically by evaluating quantiles of a Gaussian distribution with mean 0 and a standard deviation proportional to $\sigma(\Delta\tilde{\mathcal{L}})$ which decays with epoch. A full specification of the algorithm is provided in Section D.

IV. RESULTS

We now test the ability of our proposed CHILL model to recover parameters (via SEM) and node labels (via simulated annealing) on both synthetic and empirical data sets.

A. Parameter Estimation on Synthetic Data

Figure 3 illustrates the behavior of SEM in recovering parameters from synthetic hypergraphs generated by the CHILL model. Each synthetic hypergraph is generated using our model and a known parameter vector $\boldsymbol{\theta}$. We performed SEM on a set of hypergraphs which were generated with varying amounts of homophily across each of the three growth mechanisms: the edge copy step, the external node addition step, and the novel node addition step. We also varied the expected number of nodes added to e in each of these steps. To measure the success of a parameter recovery, we compute the Kullback-Leibler (KL) divergence between the true and estimated distributions described by each of the six parameters, and add the results. Using standard results, our measure of error then becomes

$$\begin{aligned} D[p_{\boldsymbol{\theta}}||\hat{p}_{\boldsymbol{\theta}}] = &\sum_{*\in\{+,-\}} c_* \left(\rho_* \log\frac{\rho_*}{\hat{\rho}_*} + (1-\rho_*)\log\frac{1-\rho_*}{1-\hat{\rho}_*}\right) \\ &+ \sum_{*\in\{+,-\}} \left(\gamma_* \log\frac{\gamma_*}{\hat{\gamma}_*} - \gamma_* + \hat{\gamma}_*\right) \\ &+ \sum_{*\in\{+,-\}} \left(\eta_* \log\frac{\eta_*}{\hat{\eta}_*} - \eta_* + \hat{\eta}_*\right)\,, \end{aligned} \tag{6}$$

where c is a factor describing, in expectation, how many nodes of each label are selected in the copy step. For the purposes of of this analysis, we set $c_* = 1$ for both labels.

As seen in Figure 3, the quality of the learned parameter estimates depends on the seeded parameter vector. We observe especially low estimation errors when the seeded parameters reflect high copy rates ρ_* and homophily as reflected in the differences $\gamma_+ - \gamma_-$ and $\eta_+ - \eta_-$; a typical SEM run for these parameter settings is shown in the top row on the right of Figure 3. That said, even in the least successful cases for estimation with high KL divergence, the learned estimates are still visually close to the true values, illustrated in the bottom row on the right of Figure 3.

B. Parameter Estimation on Empirical Data

We also used our method to infer parameters for six standard empirical hypergraph datasets described in Table I. Although many labeled hypergraph data sets are now available for community detection, relatively few have natively binary node labels. The `senate-bills` and `house-bills` datasets [37, 38] are hypergraphs of cosponsorships of bills in the U.S. Senate and House of Representatives, respectively, with node labels corresponding to the political party of each legislator; we use the version of these data sets used in [9]. The `high-school` and `primary-school` datasets [2, 39, 40] are hypergraphs of social interactions between students in a high school and primary school, respectively, with node labels corresponding to the gender of each student; these data sets were collected through the SocioPatterns project. Due to the high temporal resolution of these data, we consider a sequence of interactions to correspond to a single hyperedge of those interactions are identical and temporally contiguous. Finally, we considered a `coauthorship` data set [41] which contains publishing records for 81 computer science conferences from 2000 to 2015, with a binary gender label for each author.

Figure 4 shows our parameter estimates for each of the empirical data sets under the CHILL model. In each panel, we plot the same-label and opposite-label parameter estimates on opposing axes. Some data sets, like the gendered primary school contact network, appear on or near the line of equality in all three panels, suggesting that there is little selection in network evolution with respect to the chosen binary labels. The coauthorship data, meanwhile, falls on the line of equality for the edge copy and external node addition mechanisms, but there is a difference in the novel node addition parameters: $\hat{\eta}_+ > \hat{\eta}_-$. Our model presents a simplified picture of the process of collaboration formation as being initiated by a focal individual scientist, who then recruits from a previous collaboration, from other published scientists in the field, and from an unobserved pool of novices who have never published before. Our findings suggest that of these mechanisms, only the latter (the recruitment of novices) displays substantial gender homophily, with novices who share the gender of the focal scientist more likely to be recruited to the collaboration. This effect may be partially due to gender homophily between PhD students and their advisors, a known phenomenon in several scientific areas [43].

The `senate-bills` dataset appears to have slight homophily in the estimates of the edge copy and external node addition parameters. Because new senators

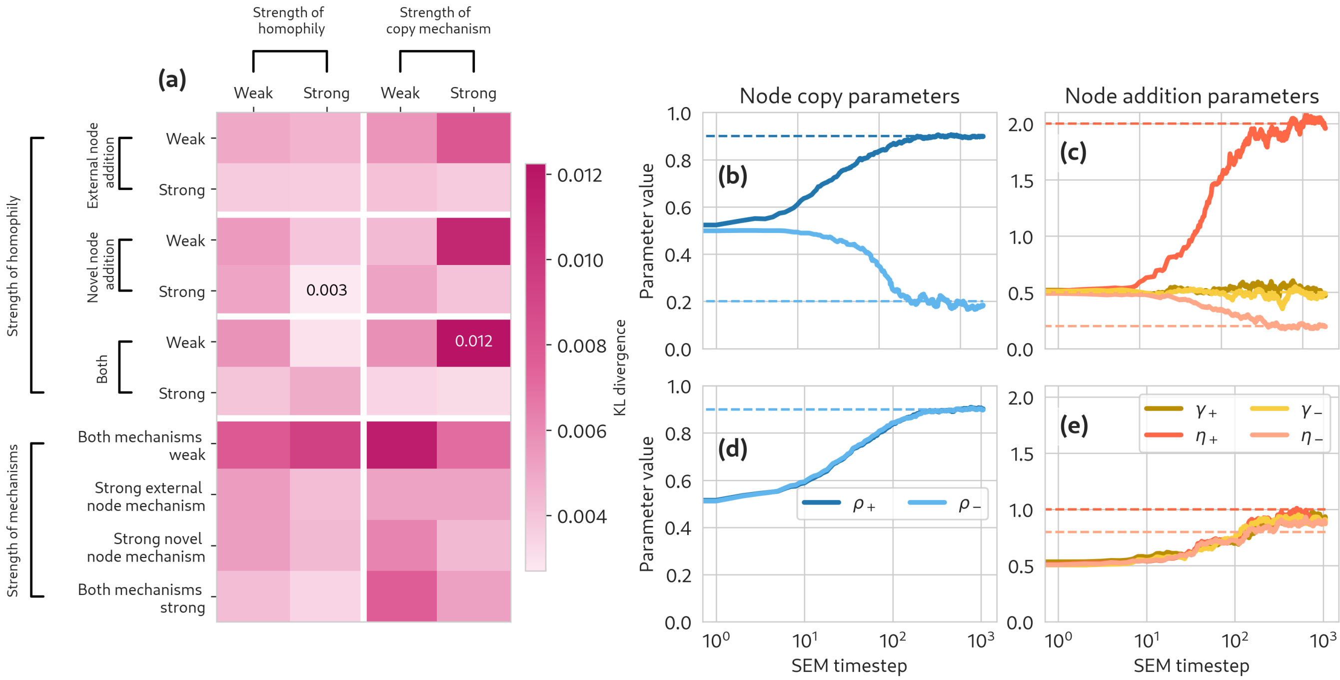

FIG. 3. Performance of SEM on synthetic hypergraphs. Each entry in the heatmap (panel (a)) corresponds to a set of parameters $\boldsymbol{\theta}$, which can be found in Appendix G. For each set of parameters, 20 independent 2000-edge hypergraphs are generated according to the model and SEM is performed independently on each. We then compute the Kullback-Leibler (KL) divergence of the estimated parameters from the true ones (eq. (6)) and average the results to obtain a value in a pixel. The parameter estimates over iterations of SEM are plotted for the underlying parameters which produced the lowest mean KL divergence (copy parameters in panel (b) and addition parameters in panel (c)), and similarly for the parameters that produced the highest mean KL divergence in panels (d) and (e).

TABLE I. Data sets used in this paper. We show the number of nodes n, edges m, average degree $\langle d \rangle$, and average edge size $\langle k \rangle$ for each dataset.

	n	m	$\langle d \rangle$	$\langle k \rangle$	Nodes are	Edges are	Labels represent
`senate-bills` [9, 37, 38]	294	29,157	789.6	8.0	Senators	Cosponsored bills	Political party
`house-bills` [9, 37, 38]	1,494	60,987	835.8	20.5	Congresspeople	Cosponsored bills	Political party
`high-school` [39, 40]	320	60,544	484.6	2.6	Students	Social interactions	Gender
`primary-school` [2]	227	44,626	590.7	3.0	Students	Social interactions	Gender
`enron-emails` [42]	14,901	83,372	18.5	3.3	Email addresses	Email threads	Fringe vs. core
`coauthorship` [41]	105,256	142,834	3.4	2.5	Authors	Papers	Gender

are elected far less frequently than bills are cosponsored, the novel node addition parameters are both nearly 0. The analogous `house-bills` data set has edge copy parameter estimates which are relatively similar to those of `senate-bills`, but which suggest a small tendency towards heterophily rather than homophily in edge-copying. The slight heterophily in the copy step of `house-bills` may be related to reciprocity in bipartisan cosponsorship, a phenomenon that has been observed in U.S. legislative bodies [1, 44]. We illustrate the connection between reciprocity and edge copy parameters through the following example. Consider some bill e, where u is a cosponsor of both bill e and bill f. If u is a Democrat, for instance, and the primary sponsor of f is a Republican, then u was a "bipartisan" cosponsor of f. Then reciprocity of bipartisan cosponsorship suggests that the Republican cosponsors of f may also cosponsor e to "return the favor." However, the other Democratic cosponsors of f are not indebted in the same way to u, perhaps making them less likely to cosponsor e.

PC: This seems helpful but I think we still owe an explanation if possible for why house and senate might be different. Possibly here? [1].

The novel node estimates are also very small. The biggest difference in our estimates for these two systems is in the external node addition parameter estimates; the `house-bills` data set has estimates $\hat{\gamma}_+$ and $\hat{\gamma}_-$ which are each roughly three times larger than their values in the `senate-bills` data set. These estimates suggest the op-

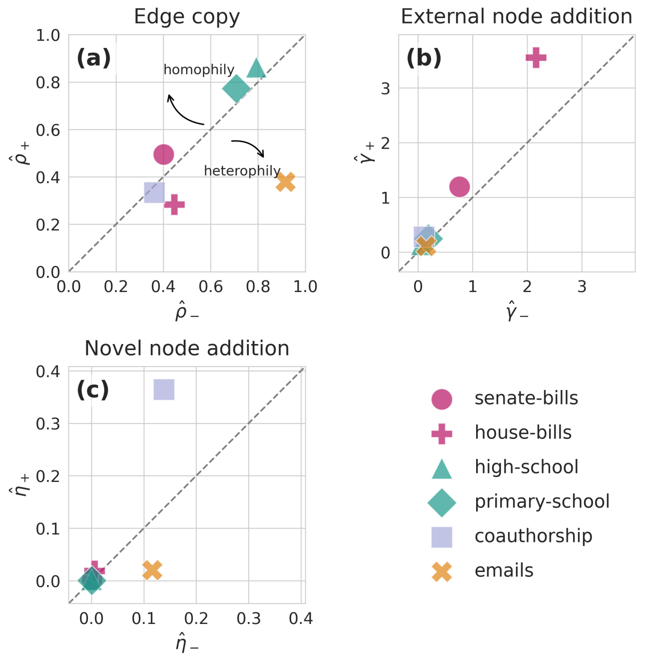

FIG. 4. Model parameters inferred via SEM for six empirical higher-order networks. The line of equality is shown as a guide for the eye.

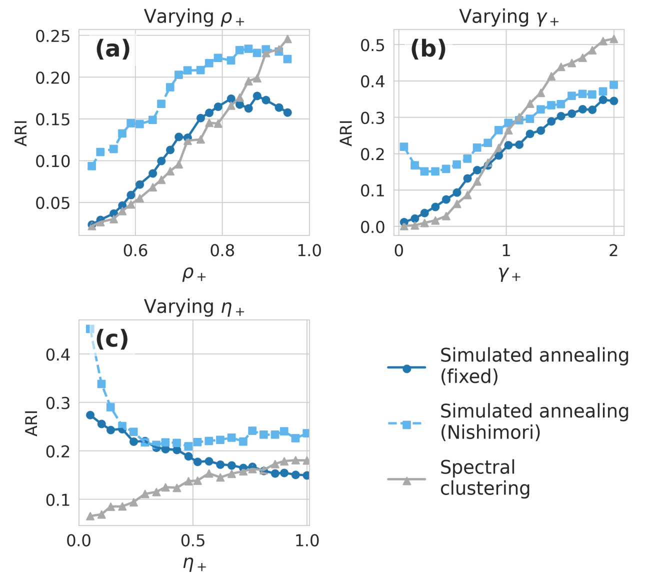

FIG. 5. Comparison of community detection methods on synthetic data. We compare our simulated annealing algorithm under the Nishimori condition and with a fixed parameter vector $\boldsymbol{\theta}' = (0.9, 0.1, 0.2, 0.1, 2.0, 0.3)$ against a baseline of Laplacian spectral clustering on the weighted clique projection of the hypergraph. The "default" parameter vector used is $\boldsymbol{\theta} = (\rho_+, \rho_-, \gamma_+, \gamma_-, \eta_+, \eta_-) = (0.8, 0.15, 0.8, 0.5, 0.7, 0.2)$. In each of the three panels (a-c), a single one of these parameters is varied while the others are held fixed at their default values. Each point gives the mean ARI over 500 independent model simulations, each of which were simulated for 200 timesteps.

eration of similar levels of party homophily in the cosponsorship of bills, but with proportionally more of the legislators sponsoring a bill originating in the House having been previously uninvolved with a previous "seeding" bill. These estimates reflect the higher mean edge size of `house-bills`, where a bill is co-sponsored by a mean of over 20 congresspersons, as opposed to the mean of 8 in `senate-bills`.

The emails hypergraph falls in the heterophilic region of the edge copy panel. This is a feature of the sampling strategy. The enron dataset contains the email data from 145 users associated with the Enron Corporation. These users are the "core" nodes of the hypergraph. Any users who were included on emails with core users, but whose complete email data was not sampled, are referred to as "fringe" nodes. Each hyperedge represents an email between a set of nodes, or users. Therefore, each hyperedge must contain at least one core node. And since 73% of edges in the hypergraph have size 2, and less than one percent of the nodes in the hypergraph are core nodes, it is likely that any given edge containins one core node and one fringe node. In fact, 67% of edges in the Enron data match exactly this description. The high magnitude of the copy parameters, particularly ρ_-, is a result of the nature of email — rarely does an email interaction between a group of users occur only once. Often, an email is precipated by an existing conversation between that set of users, making the edge copy mechanism well suited for modeling email network growth.

C. Community Detection via Simulated Annealing

We evaluate our simulated annealing algorithm on a range of small synthetic data sets under two conditions, shown in Figure 5. Under the Nishimori condition [45], we generate a synthetic hypergraph according to a parameter vector $\boldsymbol{\theta}$ and then attempt to learn $\mathbf{z}$ via simulated annealing using the same parameter vector $\boldsymbol{\theta}$. Under the "fixed" condition, which simulates a more realistic data analysis scenario, we always use the same parameter vector $\boldsymbol{\theta}'$ to learn $\mathbf{z}$ via simulated annealing, regardless of the true parameter vector $\boldsymbol{\theta}$ used to generate the synthetic data. In this condition, we use a default value of $\boldsymbol{\theta}' = (0.9, 0.1, 0.2, 0.1, 2.0, 0.3)$. We compare these methods against each other and against a baseline of Laplacian spectral clustering [46] applied to the weighted clique projection of the hypergraph. We assess the effectiveness of these techniques and our method with a measure of clustering effectiveness, the adjusted Rand index (ARI) [47]. In each panel of Figure 5, we show the performance of each of these models as a single entry of $\boldsymbol{\theta}$ is varied while the others are held fixed. Unsurprisingly, simulated annealing under the Nishimori condition outperforms simulated annealing under the fixed condition. For high levels of homophily (larger values of the varied parameter), simulated annealing under the Nishimori condition can nevertheless be outperformed by Laplacian spectral clustering. Notably, however, there are regions of parameter space, especially those with low levels of homophily (smaller values of the varied parameter) in which both simulated annealing under the fixed condition and under the Nishimori condition outperform Laplacian spectral clustering. This result is in some sense unsurprising, since we perform our experiment on synthetic samples drawn directly from the CHILL model. These experiments do suggest, however, that there exist classes of hypergraphs in which community structure can be more reliably detected via likelihood maximization under CHILL than at least this standard graph method.

We can see this phenomenon in experiments on empirical data as well. We evaluate our simulated annealing algorithm on a subset of our empirical data sets: `senate-bills`, `primary-school`, and `high-school`; the poor scaling of simulated annealing with the number of nodes n made it computationally prohibitive to run on the larger `house-bills`, `enron-emails`, and `coauthorship` data sets. We use the class membership of the students in `primary-school` and `high-school` as the ground truth labels in this section instead of gender because community structure has been previously detected with these labels [9]. Since in data analysis contexts the labels $\mathbf{z}$ (and therefore parameter estimates $\hat{\boldsymbol{\theta}}$) are not available, we instead use a single, fixed parameter vector for each of these three experiments: $\hat{\boldsymbol{\theta}} = (0.9, 0.1, 1.0, 0.25, 0.001, 0.001)$. All of these datasets are appropriately modeled with very small values of η_+ and η_- under our model due to their large number of edges compared to small number of nodes. Since in these three data sets many of the nodes exist in the system before data collection, for the purposes of community detection we ignore the novel node addition step, assuming instead that all nodes are present in the system and eligible to be added via the extant node addition step. We compare CHILL SEM against several competitors. The all-or-nothing hypergraph modularity of [9] does not allow the user to specify a desired number of clusters, but because it returns a binary partition on `senate-bills`, we can compare directly on this data set. Belief-propagation nonbacktracking spectral clustering [14] allows the user to specify the number of clusters, and the authors report results for binary clustering. We therefore compare our method against the reported results from these two articles.[1] Notably, both hypergraph modularity maximization and nonbacktracking spectral clustering are based on generative models which assume conditional edge independence, the assumption relaxed by CHILL. We also include as comparisons two dyadic methods, implemented on the weighted clique projection of the hypergraph, that allow the user to specify the number of clusters: Laplacian spectral clustering [46] and greedy modularity maximization [48].

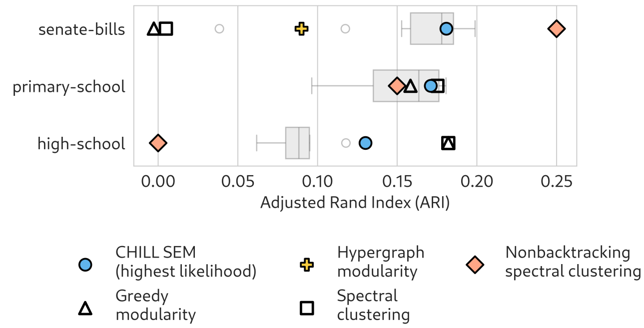

FIG. 6. Community detection results on three selected empirical datasets. Box plots describe the distribution of the ARI of CHILL SEM on the selected data set across 10 separate runs. The single label set achieving the highest likelihood under simulated annealing is highlighted. We compare these results with the ARI of the labels generated via several comparisons: hypergraph modularity [9] (only for `senate-bills`), belief-propagation nonbacktracking spectral clustering [14], greedy dyadic modularity maximization [48], and dyadic Laplacian spectral clustering [46]; the latter two dyadic methods are denoted by hollow markers. Values for hypergraph modularity maximization and nonbacktracking spectral clustering are reported in [9] and [14], respectively.

In Figure 6, we observe that CHILL simulated annealing outperforms both greedy-modularity and spectral clustering on senate bills in all 10 runs in ARI. On the primary school dataset, simulated annealing performs comparably in ARI to the competitor methods, with a slightly lower average ARI compared to spectral clustering and greedy modularity's results. However, simulated annealing performs worse on the high school dataset in ARI compared to its competitors. While the computational performance of simulated annealing presents a difficulty in its application, the detection of communities via the maximization of likelihood provides evidence that the formations of these datasets, especially senate-bills, can be simulated via our proposed model *CHILL*. Further, the success of our model in comparison to spectral clustering and greedy modularity provides evidence challenging the assumption of edge independence within `senate-bills`. This is because the baseline methods, greedy modularity and spectral clustering, are frequently justified via the stochastic block model [35, 49], which relies on the assumption of edge indpenece, where our method explicitly utilizes dependent hyperedges.

V. DISCUSSION

We have proposed Copying Hyperedges Influenced by Latent Labels (CHILL), a generative model for the for-

[1] The authors of [14] did not perform the data filtering on `high-school` and `primary-school` described in Section IV B, so our direct comparison with their results should be interpreted with some caution.

mation of binary-labeled hypergraphs in which edges form as noisy copies of previous edges, combined with the addition of extant and novel nodes. The copying, extant, and novel node addition steps are all tuned by node labels and parameters which control the strength of homophily or heterophily in each of these steps. We have demonstrated efficient parameter inference in the CHILL model via stochastic expectation maximization (SEM). We have also demonstrated that, in some circumstances, maximum-likelihood inference of node labels $\mathbf{z}$ via simulated annealing can outperform both dyadic and polyadic community detection algorithms that proceed from the assumption of conditional edge independence. We interpret this latter result as proof-of-concept that explicitly modeling edge-dependence can be a useful framework for community detection in hypergraphs.

There are several directions for future work. Most obviously, although our results show that maximum-likelihood inference in the CHILL model is promising for detecting certain kinds of community structure, the simulated annealing method we have used is impractically expensive on data sets of even relatively modest size. Developing more efficient heuristics for maximizing the likelihood over $\mathbf{z}$ will be an important direction for scalable application the CHILL model for community detection. Scalability may also imply opportunities for improved performance. We note, for example, that [14] proposed an algorithm which obtained an ARI of approximately 0.25 on the `senate-bills` data set, higher than our proposed simulated annealing method. Their algorithm requires alternating between parameter estimation and community detection phases. In principle, we could attempt a similar alternating procedure in our model, but the computational cost of simulated annealing makes this approach infeasible in practice. Future work to develop more efficient heuristics for community detection under our model could in principle enable such an alternating algorithm, with potentially improved performance.

There are also opportunities to extend CHILL to more realistic settings and more diverse data sets. One obvious extension is to nonbinary node labels, which would enable multiway community detection algorithms. Another approach is to relax the assumption that *every* edge is formed through the edge-copying mechanism; one could instead allow some edges to form spontaneously without copying an earlier edge. A final possible extension is to incorporate recency-bias [24] into the copy mechanism, so that more recent edges are more likely to be copied while edges which are sufficiently old may cease to be copied at all.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The code used to generate the results in this paper is available at `https://github.com/Violet-Ross/labeled-growth`. Data sets are publicly available through the papers cited in Table I.

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Appendix A: Power-Law Degree Distribution

Our derivation follows that of He et al. [26], which in turn is based on a derivation by Mitzenmacher [32].

At stationarity, the mean degree $\langle d \rangle$ of a node satisfies

$$\langle d \rangle = \frac{m}{n} \langle k \rangle \ , \tag{A1}$$

where $\langle k \rangle$ is the mean edge size (computable from the stationary distribution $p_{K_0,K_1}(k_1,k_2)$), m is the number of edges, and n is the number of nodes. In expectation, there are $\eta \triangleq \eta_- + \eta_+$ new nodes added per new edge, which means that $\frac{m}{n} \to \frac{1}{\eta}$ as $m, n \to \infty$ in expectation. So, we have

$$\langle d \rangle = \frac{\langle k \rangle}{\eta} \ . \tag{A2}$$

To proceed, we make a mean-field assumption that the degree d of a node is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e. Define the moments

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle \ . \tag{A3}$$

We'll first calculate the expected number of nodes of label 0 sampled in the edge-sampling step. Suppose that an edge f is sampled with K_0 nodes of label 0 and K_1 nodes of label 1, and $K_0 + K_1 = K$ total nodes. Conditional on this event, the probability that a node of label 0 is selected as the seed node u is $\frac{K_0}{K}$, and when this occurs, the expected number of nodes of label 0 in e is $1+\rho_+(K_0-1)$. If, on the other hand, a node of label 1 is selected as the seed node u, which occurs with probability $\frac{K_1}{K}$, then the expected number of nodes of label 0 in e is $\rho_- K_0$. So, the expected number of nodes of label 0 copied into the new edge e from f is

$$\begin{aligned} \sigma_0 &\triangleq \left\langle \frac{k_0}{k}\left(1+\rho_+(k_0-1)\right) + \frac{k_1}{k}\rho_- k_0 \right\rangle && \text{(A4)} \\ &= 1 + \rho_+\left[\rho_{00}+\rho_{11}-1\right] + \rho_-\rho_{01} \ . && \text{(A5)} \end{aligned}$$

Similarly, the expected number of nodes of label 1 copied into the new edge e from f is

$$\begin{aligned} \sigma_1 &\triangleq \left\langle \frac{k_1}{k}\left(1+\rho_+(k_1-1)\right) + \frac{k_0}{k}\rho_- k_1 \right\rangle && \text{(A6)} \\ &= 1 + \rho_+\left[\rho_{00}+\rho_{11}-1\right] + \rho_-\rho_{01} \ . && \text{(A7)} \end{aligned}$$

We let

$$\sigma \triangleq \sigma_0 + \sigma_1 = 1 + \rho_+(\rho_{00}+\rho_{11}-1) + 2\rho_-\rho_{01} \ . \tag{A8}$$

Importantly, σ depends only on the parameters and the moments of the joint distribution $p_{K_0,K_1}(k_1, k_2)$, which can be computed via linear algebra as described in Section II A 1.

We now invoke the mean-field assumption. Under this assumption, the probability that a single node selected in the sampling step has degree d is $dp_d^{(t)}$, where $p_d^{(t)}$ is the fraction of nodes with degree d at time t, since that node participates in d edges, independently of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e. The expected number of nodes of degree d sampled in the edge-sampling step is then $\sigma \frac{dp_d^{(t)}}{\langle d \rangle}$, and the expected *change* in the count of degree d nodes through the edge-sampling step is

$$\frac{\sigma}{\langle d \rangle}((d-1)p_{d-1}^{(t)} - dp_d^{(t)}) \tag{A9}$$

Similarly, the expected number of nodes which are sampled in the extant node addition step is $\gamma \triangleq \gamma_+ + \gamma_-$. Since this step does not involve any selection proportional to degree, the expected change in the count of degree d nodes through this step is

$$n^{(t+1)}p_d^{(t+1)} - n^{(t)}p_d^{(t)} = \frac{\sigma}{\langle d \rangle}\left[(d-1)p_{d-1}^{(t)} - dp_d^{(t)}\right] + \gamma(p_{d-1}^{(t)} - p_d^{(t)}) \,. \tag{A10}$$

We know that at stationarity, $n^{(t+1)} - n^{(t)}$ is in expectation equal to $\eta \triangleq \eta_+ + \eta_-$, the expected number of novel nodes, while at stationarity $p_d^{(t+1)} = p_d^{(t)} = p_d$ independent of t. Thus, we obtain

$$p_d \eta = \frac{\sigma}{\langle d \rangle}\left[(d-1)p_{d-1} - dp_d\right] + \gamma(p_{d-1} - p_d) \,.$$

Rearranging, we obtain

$$\begin{aligned}\frac{p_d}{p_{d-1}} &= \frac{\frac{\sigma}{\langle d \rangle}(d-1) + \gamma}{\frac{\sigma}{\langle d \rangle}d + \eta + \gamma} \\ &= 1 - \frac{\frac{\sigma}{\langle d \rangle} + \eta}{\frac{\sigma}{\langle d \rangle}d + \eta + \gamma} \,.\end{aligned}$$

Allowing d to become large (which describes the tail of the degree distribution), we have

$$\begin{aligned}\frac{p_d}{p_{d-1}} &\approx 1 - \frac{1}{d}\frac{\frac{\sigma}{\langle d \rangle} + \eta}{\frac{\sigma}{\langle d \rangle}} \\ &= 1 - \frac{1}{d}\left(1 + \frac{\eta}{\sigma / \langle d \rangle}\right) \\ &= 1 - \frac{1}{d}\left(1 + \frac{\langle k \rangle}{\sigma}\right) \,.\end{aligned}$$

a relation which corresponds to a power law with exponent $\zeta = 1 + \frac{\langle k \rangle}{\sigma}$. Expanding out σ yields Equation (2) in the main text. Explicitly calculating this exponent requires the moments $\langle k \rangle$, ρ_{00}, ρ_{11}, and ρ_{01}, which can be computed from the stationary joint distribution $p_{K_0,K_1}^{(\infty)}(k_1, k_2)$.

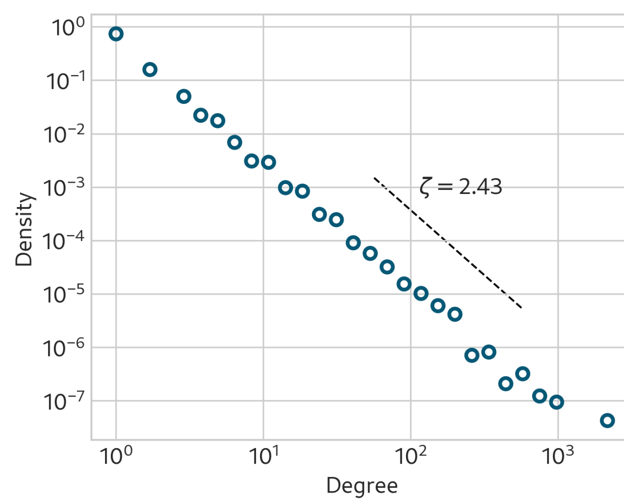

FIG. 7. Degree distribution from model simulation of 10^4 edges (points) and theoretical power law (line) with exponent given by Equation (2). The parameters in this experiment are $\rho_+ = 0.7, \rho_- = 0.3, \gamma_+ = 0.5, \gamma_- = \eta_- 3, \eta_+ = 0.2, \eta_- = 0.1$.

Appendix B: More on Stochastic Expectation-Maximization (SEM)

1. Sufficient Statistics

We describe the function σ from Section III A which computes the conditional sufficient statistics for the parameters $\boldsymbol{\theta} = (\rho_+, \rho_-, \gamma_+, \gamma_-, \eta_+, \eta_-)$ given an observed edge e, a candidate parent edge f, a candidate seed node u, and the node labels $\mathbf{z}$. Let $z = z_u$ be the label of the seed node u. Let e_z be the set of nodes in e with label z, and let f_z be the set of nodes in f with label z. Then, the sufficient statistics for the parameters $\boldsymbol{\theta}$ are given by

$$\sigma(e,f,u,\mathbf{z}) = \begin{pmatrix} |e_z \cap f_z| \\ |f_z| \\ |e_{\bar{z}} \cap f_{\bar{z}}| \\ |f_{\bar{z}}| \\ \left|(e_z \setminus f_z) \cap \mathcal{V}^{(t)}\right| \\ \left|(e_{\bar{z}} \setminus f_{\bar{z}}) \cap \mathcal{V}^{(t)}\right| \\ \left|e_z \setminus \mathcal{V}^{(t)}\right| \\ \left|e_{\bar{z}} \setminus \mathcal{V}^{(t)}\right| \end{pmatrix}^\top .$$

As descrbed in Section III A, the vector $\mathbf{s}$ of expected sufficient statistics is computed as a weighted average of $\sigma(e,f,u,\mathbf{z})$ over the belief distribution $p(f,u|e,\hat{\boldsymbol{\theta}}^{(\ell)})$ over the latent variables f and u. Then, the function g which computes the maximum likelihood estimates of the parameters $\boldsymbol{\theta}$ from $\mathbf{s}$ is given by

$$g(\mathbf{s}) = \left(\frac{s_1-1}{s_2-1}, \frac{s_3}{s_4}, s_5, s_6, s_7, s_8\right) .$$

2. Implementation Details

We use an exponentially decaying learning rate for SEMwith form $\alpha^{(t)} = \alpha^{(0)} \cdot e^{-ct}$. The initial learning rate is $\alpha^{(0)} = 0.01$ and the exponent is $c = 0.001$. We initialized the vector of sufficient statistics $\mathbf{s}^{(0)} = (1, 2, 1, 2, 0.5, 0.5, 0.5, 0.5)$. Other reasonable choices of hyperparameters did not substantially change the results of our experiments.

Appendix C: Approximation Argument for Simulated Anealing

This section provides further detail on the approximation to the likelihood described in Section III B and used in our simulated annealing implementation. The likelihood of an edge e being generated in timestep t can be written

$$\mathbb{P}(e|\mathbf{z};\boldsymbol{\theta}) = \sum_{\substack{f \prec e \\ u \in f}} \mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta}) \tag{C1}$$

$$= \frac{1}{m_t} \sum_{f \prec e} \frac{1}{|f|} \sum_{u \in f \cap e} \mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta}) , \tag{C2}$$

where m_t is the number of edges at time t. It's helpful to reindex this sum by the size of the intersection $|f \cap e|$ between f and e.

$$\mathbb{P}(e|\mathbf{z};\boldsymbol{\theta}) = \sum_{h=0}^{|e|} \sum_{\substack{f \prec e \\ |f \cap e| = k}} \frac{1}{m_t} \sum_{u \in f \cap e} \frac{1}{|f|} \mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta}) . \tag{C3}$$

We'll now argue that, when n is large, the terms in this sum corresponding to the largest intersections (for which $h = \mathrm{argmax}_{f \prec e} |f \cap e|$) will dominate the likelihood, allowing us to neglect edges f whose intersection with e is smaller.

Fix edge e and candidate parent edge f. We subscript e_z, f_z, and $\mathcal{V}_z$ to denote the subsets of e, f, and $\mathcal{V}$ with label z. Let $h_z = |(e_z \setminus f_z) \cap \mathcal{V}_z|$ be the number of extant nodes of label z in e. Then, the extant node addition process appears in $\mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta})$ as a factor

$$E(h_0,h_1) = \frac{\mathrm{Poisson}(h_{z_u}|\gamma_+) \times \mathrm{Poisson}(h_{\bar{z}_u}|\gamma_-)}{\binom{|\mathcal{V}_0^t| - |f_0|}{h_0}\binom{|\mathcal{V}_1^t| - |f_1|}{h_1}} , \tag{C4}$$

where the numerator describes the likelihood of adding fixed numbers of extant nodes of each label and the denominator describes the likelihood of selecting the *specific set* of extant nodes from the set of all candidates. Under the assumption that $n \gg |f|, |e|$, we can estimate

$$\binom{|\mathcal{V}_z^t| - |f_z|}{h_z} \approx \frac{(|\mathcal{V}_z^t|)^{h_z}}{h_z!} \approx \frac{(c_z n)^{h_z}}{h_z!} , \tag{C5}$$

where we have defined $c_z = \frac{|\mathcal{V}_z^t|}{n}$ as the fraction of nodes with label z. Then, absorbing terms that do not depend on n into a constant term $C(h_0,h_1)$, we have

$$E(h_0,h_1) = C(h_0,h_1) n^{-h_0-h_1} + O(n^{-h_0-h_1-1}) \tag{C6}$$

$$= C(h_0,h_1) n^{-h} + O(n^{-h-1}) , \tag{C7}$$

where $h = h_0 + h_1 = |(e \setminus f) \cap \mathcal{V}|$ is the total number of extant nodes added to e and

$$C(h_0,h_1) = \frac{\mathrm{Poisson}(h_{z_u}|\gamma_+) \times \mathrm{Poisson}(h_{\bar{z}_u}|\gamma_-)}{\frac{(c_0)^{h_0}}{h_0!}\frac{(c_1)^{h_1}}{h_1!}} . \tag{C8}$$

It follows that $\mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta})$ is of order $O(n^{-h})$ as n grows large, where h is the number of extant nodes added to e. Terms corresponding to choices of f for which h is larger (of which there may be up to approximately m) decay quickly for large n. We therefore have

$$\mathbb{P}(e|\mathbf{z};\boldsymbol{\theta}) = \sum_{\substack{f \prec e \\ |f \cap e| = h'}} \frac{1}{m_t} \sum_{u \in f \cap e} \frac{1}{|f|} \mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta}) + mO(n^{-h'-1}) \tag{C9}$$

where $h' = \min_{f \prec e} |(e \setminus f) \cap \mathcal{V}|$ is the minimum number of extant nodes added to e across all choices of f.

This argument suggests that, when n is relatively large in relation to m, the likelihood of e is dominated by terms corresponding to choices of f for which the number of extant nodes added to e is small. There are two limitations to this argument in the context of empirical data sets. First, it may be the case that only a small number of candidate parent edges f have small h, which may lead to a poor approximation of the likelihood when we evaluate only a small number of corresponding terms. Second, several of our data sets have m large in relation to n, making it dubious to neglect the error term. We therefore introduce two variations to our approximation heuristic:

- **Top j heuristic**: Rather than evaluate only terms for f for which h is minimal, we evaluate terms for the top j choices of f with the smallest h.

- **Batch normalization**: Rather than normalizing by $\frac{1}{m_t}$ in eq. (C9), we normalize by the number of f choices we evaluate for a specific e. This may be j under the top j heuristic, or it may be the number of f choices with h equal to the minimum h' if we do not use the top j heuristic. Letting $\mathcal{F}_e$ be the set of f choices we evaluate for a specific e, our approximation is

$$\tilde{\mathbb{P}}(e|\mathbf{z};\boldsymbol{\theta}) = \frac{1}{|\mathcal{F}_e|} \sum_{f\in\mathcal{F}_e} \sum_{u\in f\cap e} \frac{1}{|f|} \mathbb{P}(e|f,u,\mathbf{z};\boldsymbol{\theta}) \ . \tag{C10}$$

Meaning the approximate likelihood of the entire $\mathcal{H}$ is

$$\tilde{\mathbb{P}}(\mathcal{H},\mathbf{z};\boldsymbol{\theta}) = \prod_{e\in\mathcal{E}} \tilde{\mathbb{P}}(e|\mathbf{z};\boldsymbol{\theta}) \tag{C11}$$

It is important to note that the batch normalization eq. (C11) tends to overestimate the likelihood, since we have restricted ourselves to an average of high-probability seed-edges. For the purposes of simulated annealing, this is not a major problem since we are only interested in the relative likelihood of different labelings $\mathbf{z}$ and not the absolute likelihood.

Appendix D: Simulated Annealing for Community Detection

We describe the simulated annealing algorithm implementation in detail. Define the number of epochs for the simulated annealing algorithm $A = 20$. Let B be the probability that the simulated annealing algorithm accepts a change on labels that lowers the approximated likelihood

Algorithm 2 `SimulatedAnnealing`$(\mathcal{H}, \theta)$

Require: Hypergraph $\mathcal{H}$, parameters $\boldsymbol{\theta}$, number of epochs A
Ensure: Label set $\mathbf{z}$, approximation of the true labels $\mathbf{z}^*$

```
Sample z ~ Uniform({0,1})^n
ΔL ← ∅
σ_H ← 0
for t = 0 to A do
    for τ = 0 to n do
        Sample j ~ Uniform({0, ..., n − 1})
        z' ← z
        z'_j ← 1 − z'_j
        ΔE ← log P̃(H, z'|θ) − log P̃(H, z|θ)
        if t = 0 then
            ΔL ← ΔL ∪ {ΔE}
        if σ_H = 0 & t > 0 then
            σ_H ← σ(ΔL)
        if ΔE > 0 then
            z ← z'
        else
            B ← 1
            ζ_a ← ΔE/σ_H
            if 1 ≤ t ≤ 4 then
                B ← Normal(ζ_a | μ = 0, σ = 2)
            else if 5 ≤ t then
                B ← Normal(ζ_a | μ = 0, σ = 2(1 − t/A))
            Sample x ~ Uniform(0, 1)
            if B > x then
                z ← z'
return z
```

Here,

$$\sigma(\Delta\mathcal{L}) = \sqrt{\frac{\sum_{\delta\in\Delta\mathcal{L}} \delta^2 - (\frac{1}{|\Delta\mathcal{L}|}\sum_{\delta\in\Delta\mathcal{L}} \delta)^2}{|\Delta\mathcal{L}| - 1}} \tag{D1}$$

Appendix E: Simulated Annealing Illustration

In this simulation, we generated a simple hypergraph according to our model with 61 nodes and 200 edges. Running simulated annealing for 1220 steps, we obtained a maximum log-likelihood of -9.49 nats per edge with an Adjusted Rand Index (ARI) against ground-truth of 0.18. Greedy modularity maximization as implemented in the Python package `networkx` [50] obtained an ARI of 0.04. The parameters for the model used to generate the hypergraph are

$$\begin{aligned} \rho_+ &= 0.90 \\ \rho_- &= 0.10 \\ \gamma_+ &= 1.00 \\ \gamma_- &= 0.25 \\ \eta_+ &= 0.20 \\ \eta_- &= 0.10 \ . \end{aligned}$$

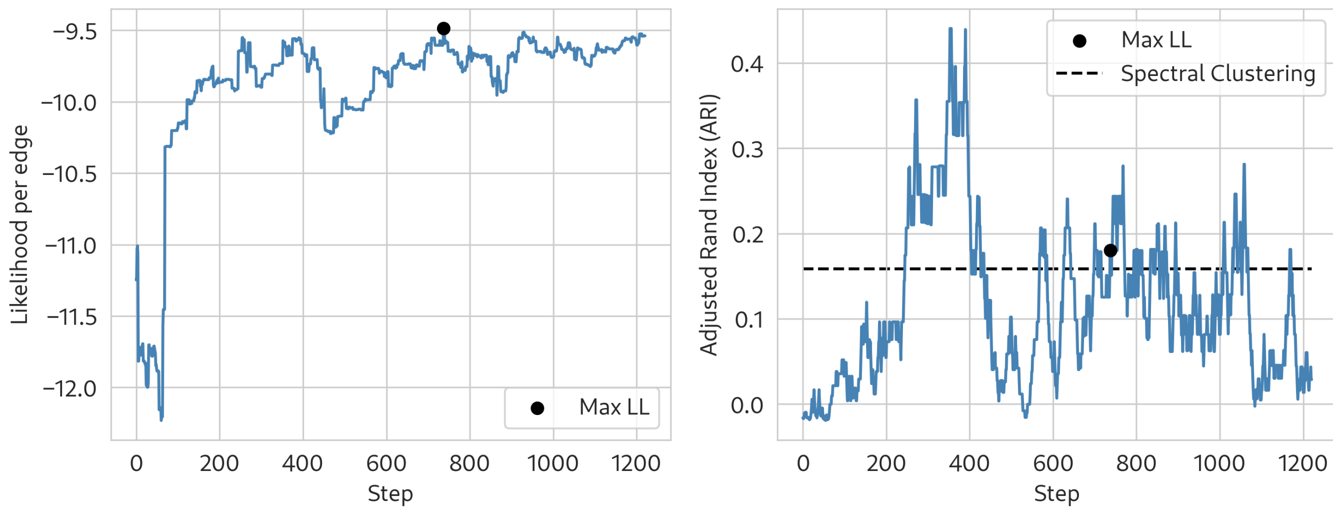

FIG. 8. Illustrative run of simulated annealing for community detection in a hypergraph generated from our model. Dashed line on the right shows the ARI achieved by Laplacian spectral clustering applied to the weighted dyadic graph projection.

The guessed parameters used to compute the likelihood during simulated annealing are

$$\begin{aligned}\hat{\rho}_+ &= 0.90\\ \hat{\rho}_- &= 0.10\\ \hat{\gamma}_+ &= 1.00\\ \hat{\gamma}_- &= 0.25\\ \hat{\eta}_+ &= 0.20\\ \hat{\eta}_- &= 0.10\,.\end{aligned}$$

Appendix F: More Simulated Annealing on Synthetic Data

Appendix G: Synthetic SEM Parameter Values

TABLE II. Strength of Homophily in Edge Copying

	ρ_+	ρ_-
Weak	0.6	0.4
Strong	0.9	0.2

TABLE III. Strength of Homophily in External and Novel Node Addition

		γ_+	γ_-	η_+	η_-
External node addition	Strong	1.0	0.8	0.2	0.2
	Weak	2.0	0.2	0.2	0.2
Novel node addition	Strong	0.2	0.2	1.0	0.8
	Weak	0.2	0.2	2.0	0.2
Both	Strong	1.0	0.8	1.0	0.8
	Weak	2.0	0.2	2.0	0.2

TABLE IV. Strength of Copy Mechanism

	ρ_+	ρ_-
Weak	0.1	0.1
Strong	0.9	0.9

TABLE V. Strength of External and Novel Node Addition Mechanisms

	γ_+	γ_-	η_+	η_-
Both weak	0.1	0.1	0.1	0.1
Strong external node mechanism	2.0	2.0	0.2	0.2
Weak novel node mechanism	0.2	0.2	2.0	2.0
Both strong	2.0	2.0	2.0	2.0